Project Ideas

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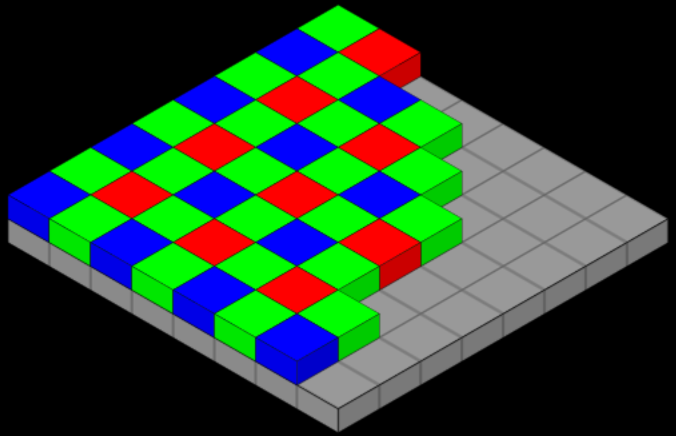
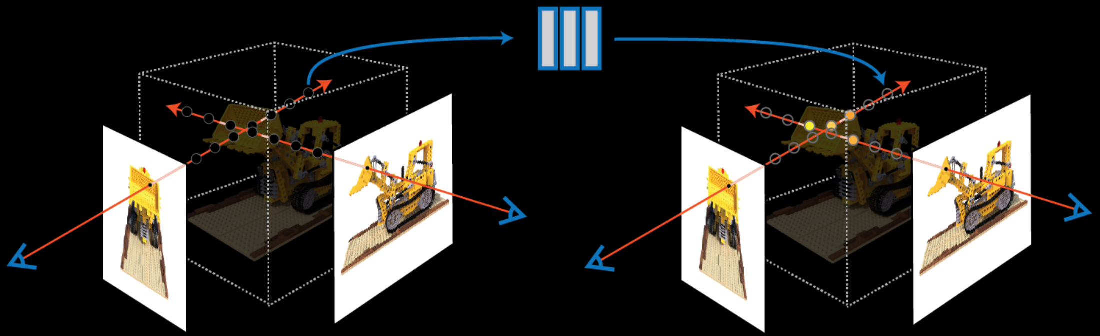
List of Projects

1. RGB Demosaicking + neural radial Field (NERF)
2. 3D point Cloud and Gaussian splatting
3. Event based and Gaussian Splatting
4. Single pixel camera + Gaussian splatting
5. Snapshot video imaging + Gaussian splatting
6. Multispectral demosaicking + filter design
7. Multispectral deconvolution

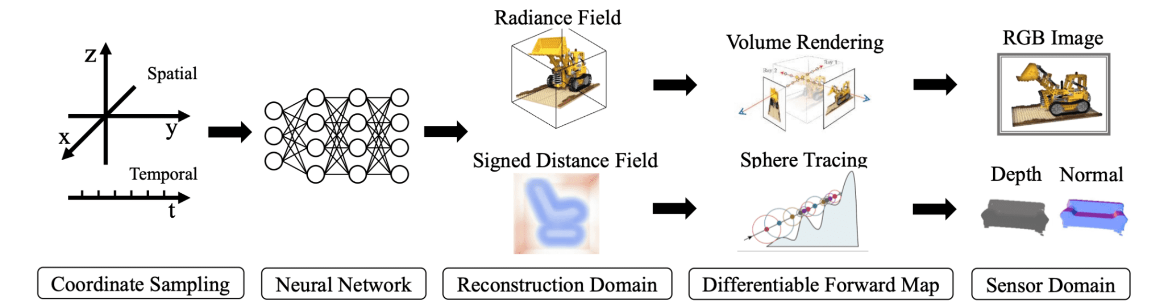
RGB Demosaicking + Neural Radial Fields



Bayer filter



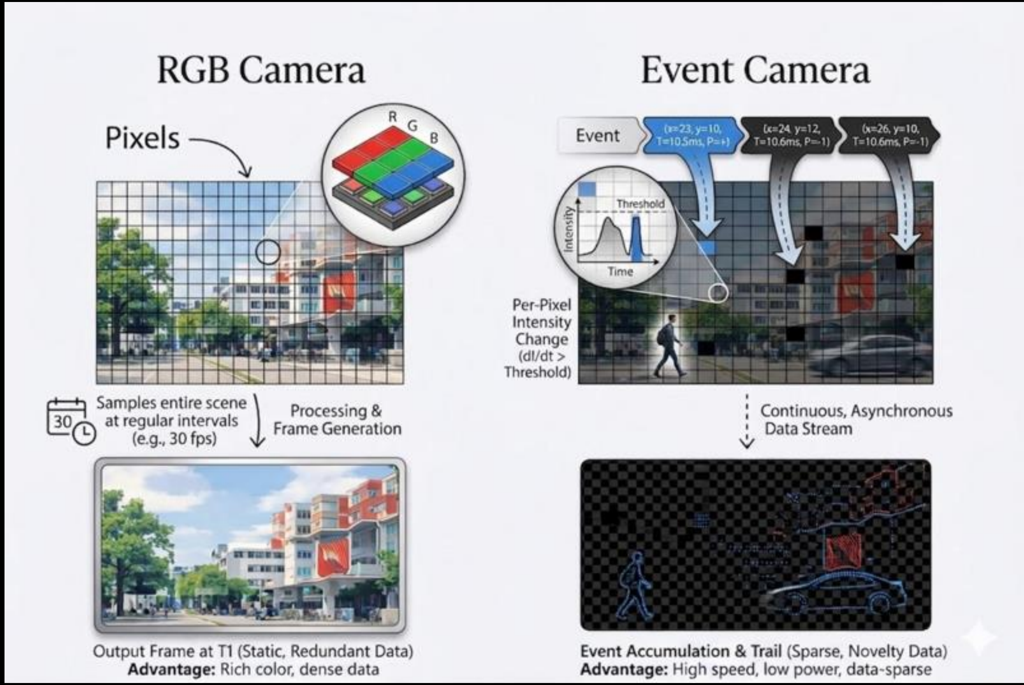
Neural Radial Fields (NERF)



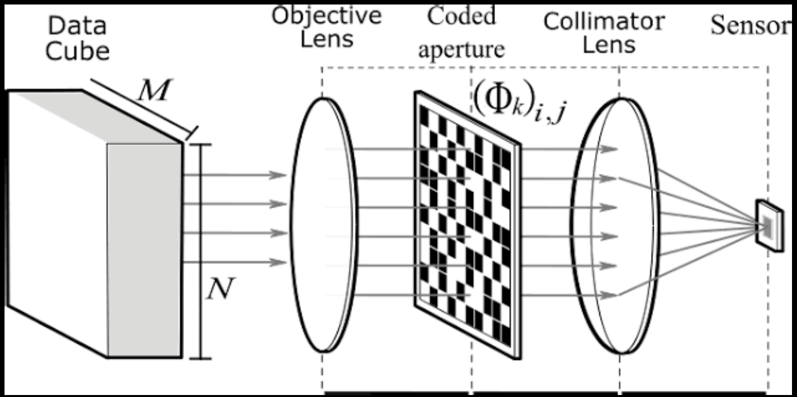
NERF requires multiple view angles

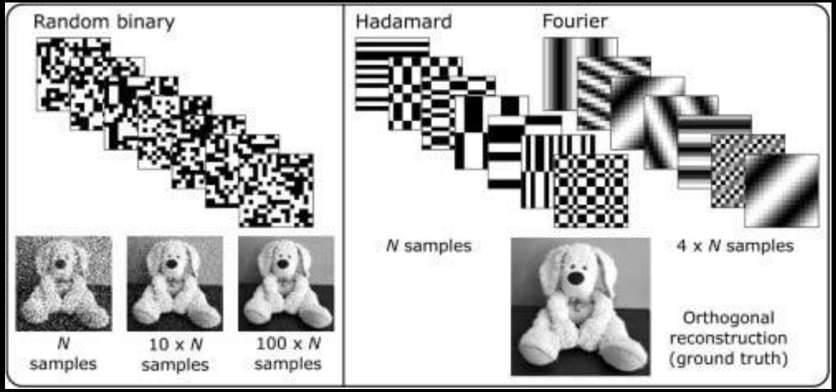
Point Cloud + Gaussian Splatting

Event Camara+ Gaussian Splatting



Single Pixel Camera + Gaussian splatting

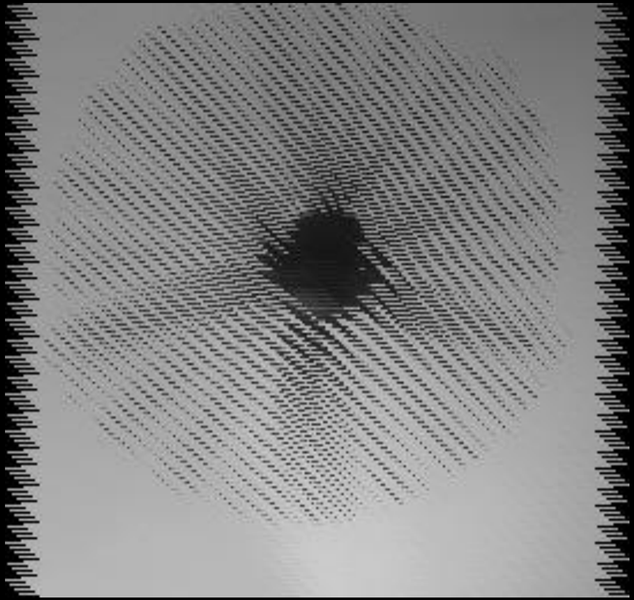


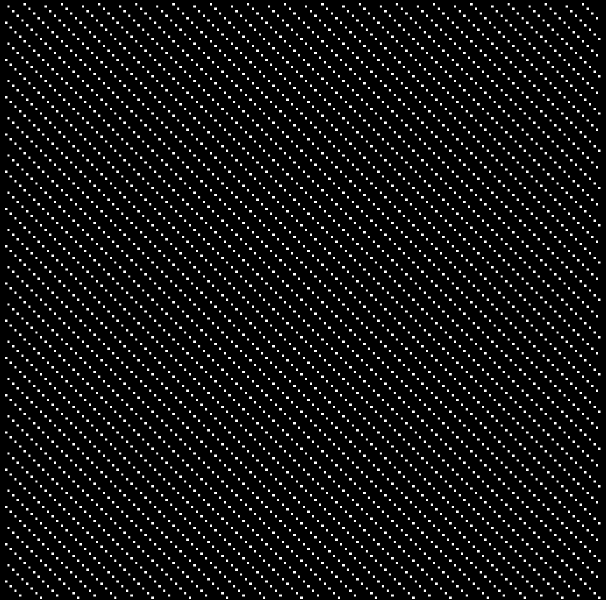


Snapshot video + Gaussian splatting

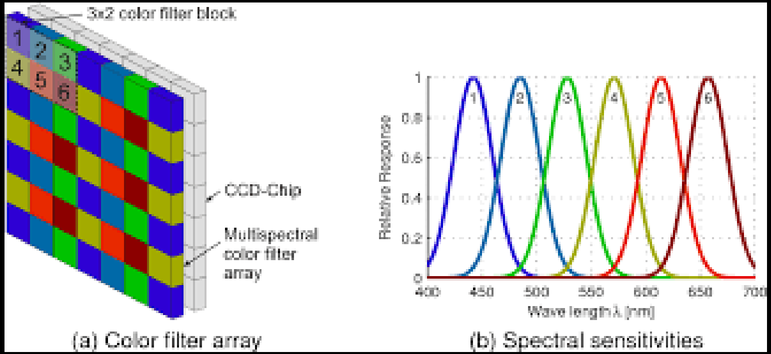
$$Y_{(:,\bar{t})} = \sum_{t=0}^{T-1} X_{(:,:,\bar{t})} \odot C_{(:,:,\bar{t})} + \Omega$$

novel aperture design

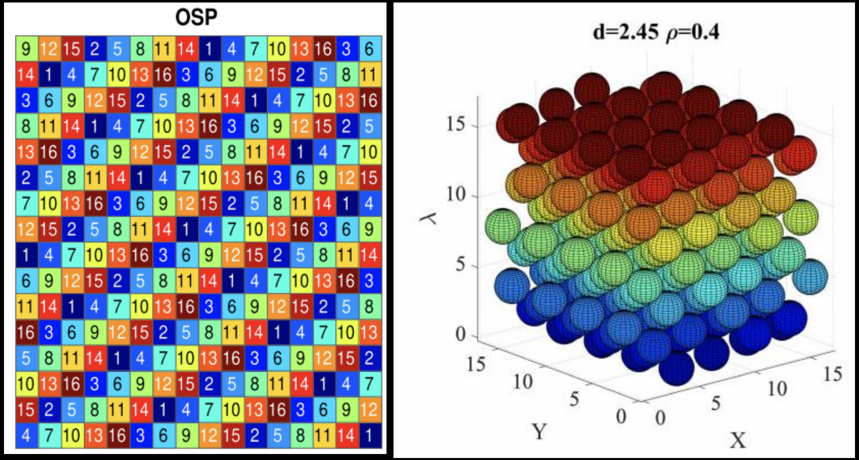




Multispectral Filter Array + Sphere Packing Bilinear Demosaicking for any Number of Filter



Multispectral Filter Array



Sphere packing
Multispectral Filter Array

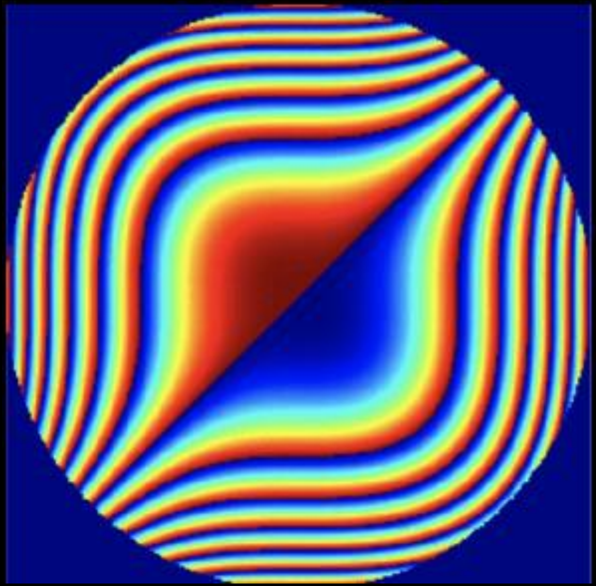
Goal: To find the deconvolution kernel for any number of Filter?

Deconvolution using neural networks

$$Y_{RGB,z} = S_{RGB}(PSF_{\lambda,z} \circledast X_{\lambda,z}) + \Omega_{\lambda,z}$$

designing the encoding is critical
to facilitate the inverse problem

measurement



DOE

Scene